Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

a. Participant name: YASMEEN BEGUM
b. Problem Statement: Build and deploy AI agents using Gemini, ADK, and Cloud Run

Brief about the idea

The **Hotel Food Summarizer Agent** is an AI workflow designed to help users quickly learn about hotel food traditions and breakfast options around the world. It works by taking a user's query, saving it into state, and then using a researcher agent to gather relevant information from Wikipedia. That research is passed to a formatter agent, which synthesizes the findings into a concise, conversational summary. The idea is to combine external knowledge with the summarization power of Gemini-2.5-flash, so the agent can provide clear, engaging answers about cuisines, ingredients, and cultural practices. Deployed on Cloud Run with a developer UI, it allows users to type queries directly and receive polished responses, making it useful for hospitality training, travel guides, or anyone curious about global food culture.

Meeting the Build Criteria

- How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

I navigated the build criteria by choosing the **Agent Development Kit (ADK)** track and designing a modular agent workflow. I used a root agent to capture user prompts, a researcher agent to query Wikipedia, and a formatter agent powered by Gemini-2.5-flash to synthesize results. This structure ensured compliance with the “What You Must Build” requirement by integrating external knowledge and producing clear summaries. Finally, I deployed the agent to Cloud Run with a Dev UI, making it reproducible, interactive, and aligned with the lab’s custom solution goals.

Opportunities

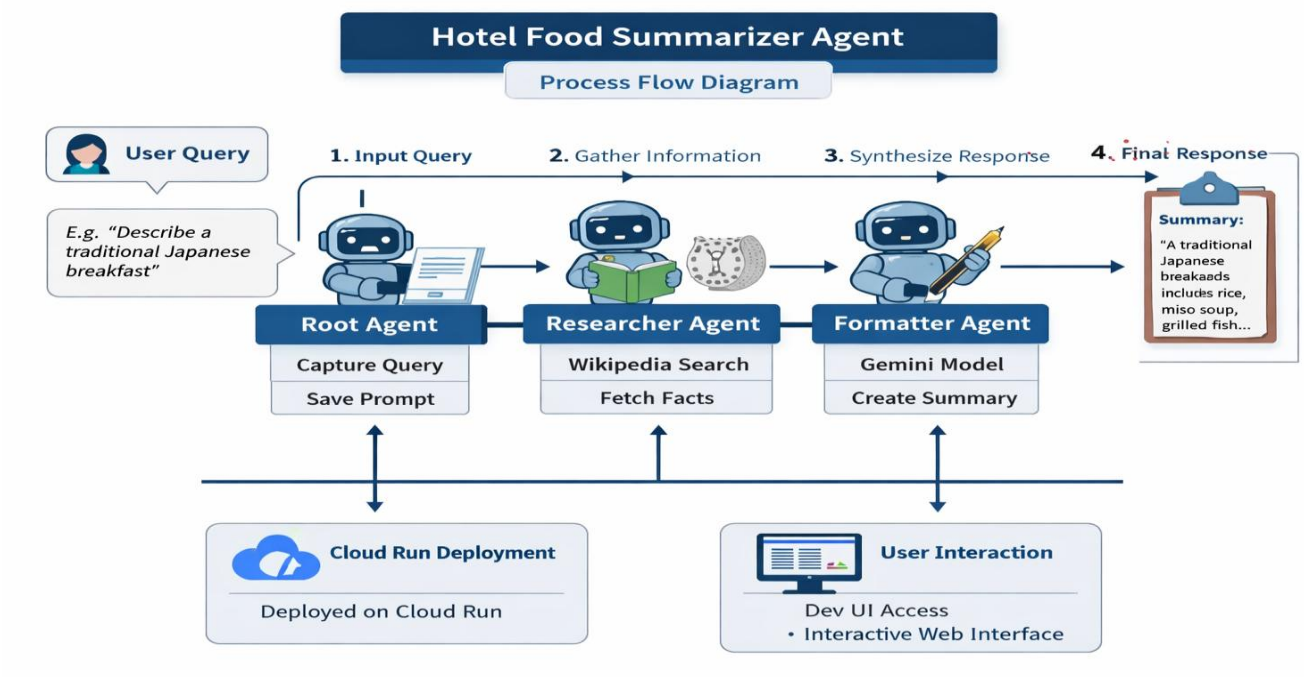
- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

•**Difference from existing ideas:**
This agent is unique because it focuses specifically on hotel food and breakfast traditions, combining Wikipedia research with Gemini summarization in a modular workflow.
•**How it solves the problem:**
It addresses the challenge of scattered food information by delivering concise, engaging summaries that are easy to understand and useful for hospitality or cultural learning.
•**USP of the solution:**
Its strength lies in seamless integration of ADK orchestration, LangChain tools, and Gemini models, deployed on Cloud Run with a Dev UI for scalability and direct user interaction.

List of features offered by the solution

- **Multi-agent workflow**: Root agent captures queries, researcher agent gathers facts, and formatter agent produces concise summaries.
- **External knowledge integration**: Uses Wikipedia search to pull reliable food and cuisine information.
- **LLM summarization**: Synthesizes research into clear, engaging responses with Gemini-2.5-flash.
- **Cloud Run deployment**: Fully containerized and deployed with a public service URL for easy access.
- **Developer UI**: Interactive web interface (/dev-ui/) for typing queries directly and viewing responses.
- **Scalability & reproducibility**: Modular design with environment variables and service accounts for compliance and repeatable deployment.
- **Custom labeling & metadata**: Supports service labels (e.g., category=food) for organization and tracking.
- **Interactive trace view**: Shows step-by-step agent calls, tool usage, and final outputs for debugging and transparency.

Process flow diagram or Use-case diagram



Technologies to be used in the solution

- **Agent Development Kit (ADK)** – for building modular, multi-agent workflows.
- **LangChain Wikipedia Tool** – to fetch external knowledge from Wikipedia.
- **Gemini-2.5-flash (Vertex AI)** – for summarizing research into clear responses.
- **Cloud Run** – to deploy the agent as a scalable web service.
- **Dev UI (ADK)** – for direct user interaction via a browser interface.
- **Python + uv + virtualenv** – for local development and dependency management.
- **IAM & Service Accounts** – for secure access to Vertex AI and deployment services.
- **Environment Variables (.env)** – to configure model, project, and region settings.

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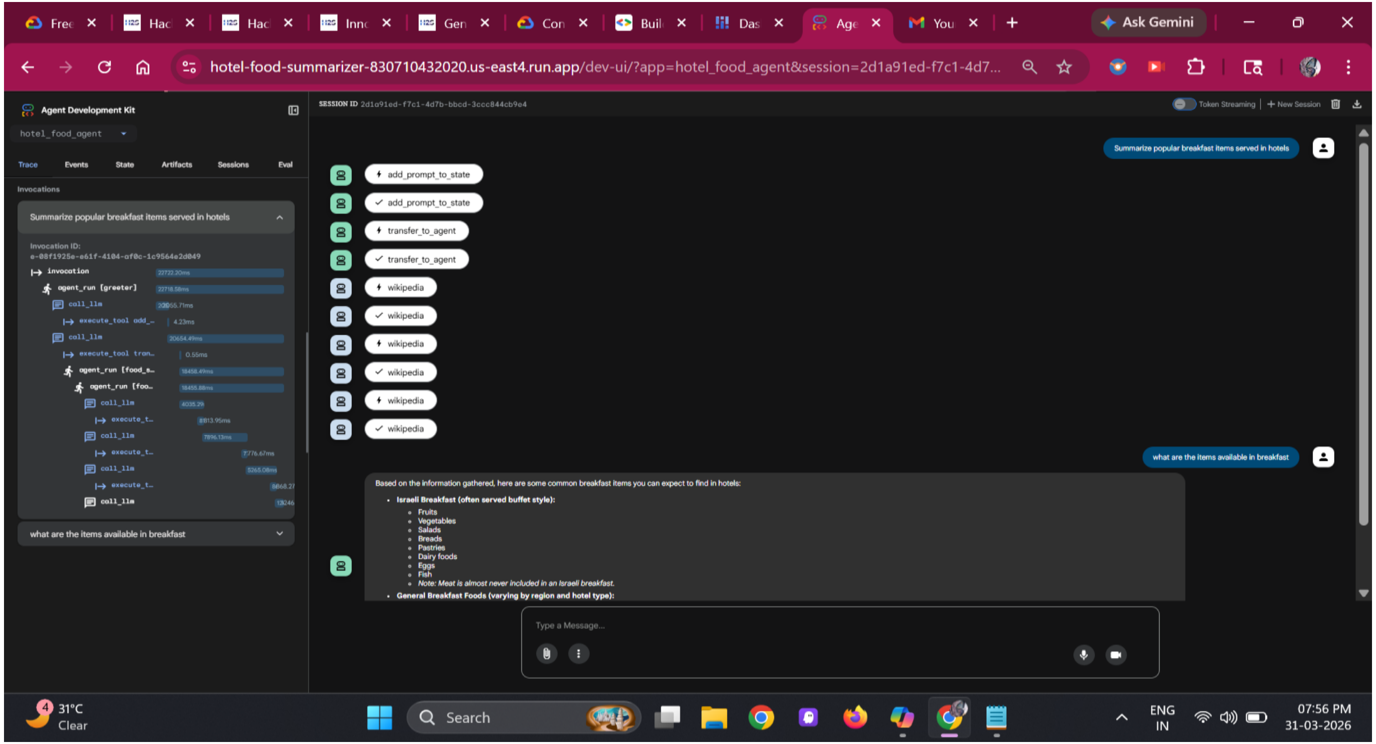
Snapshots of the prototype



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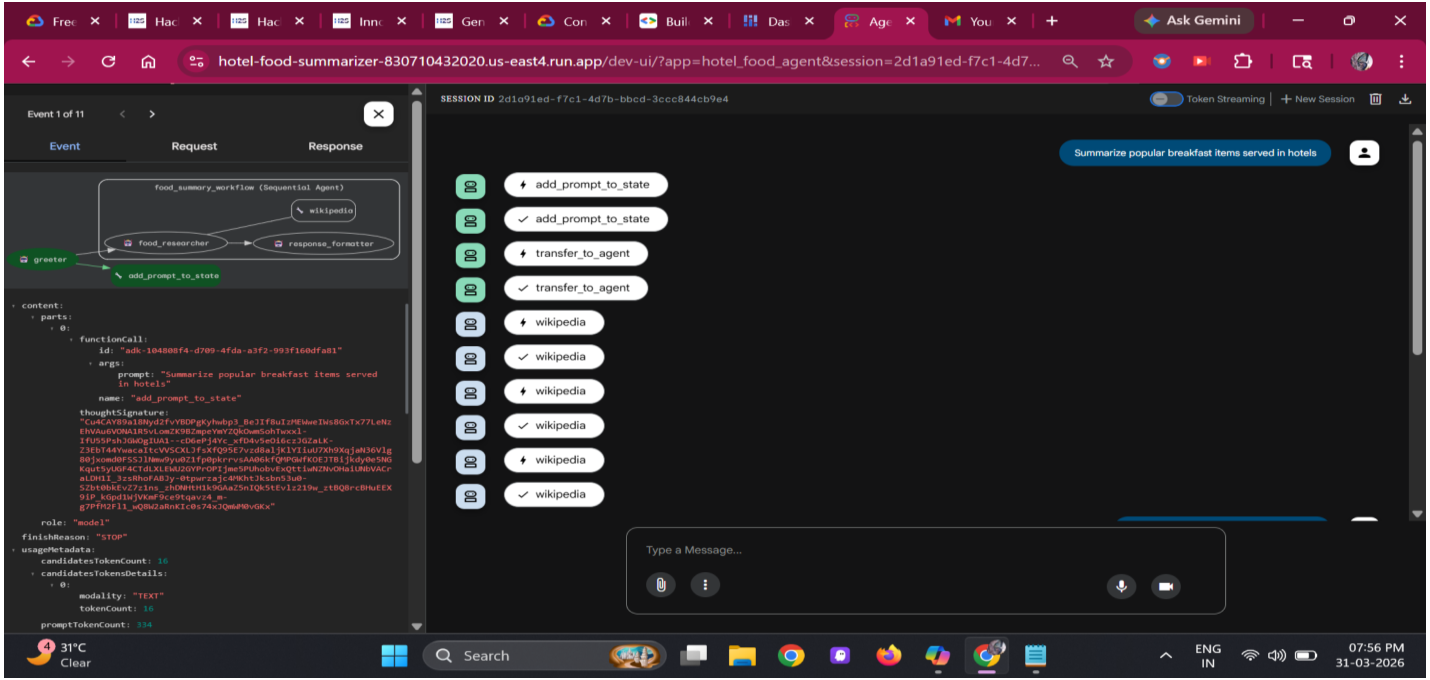

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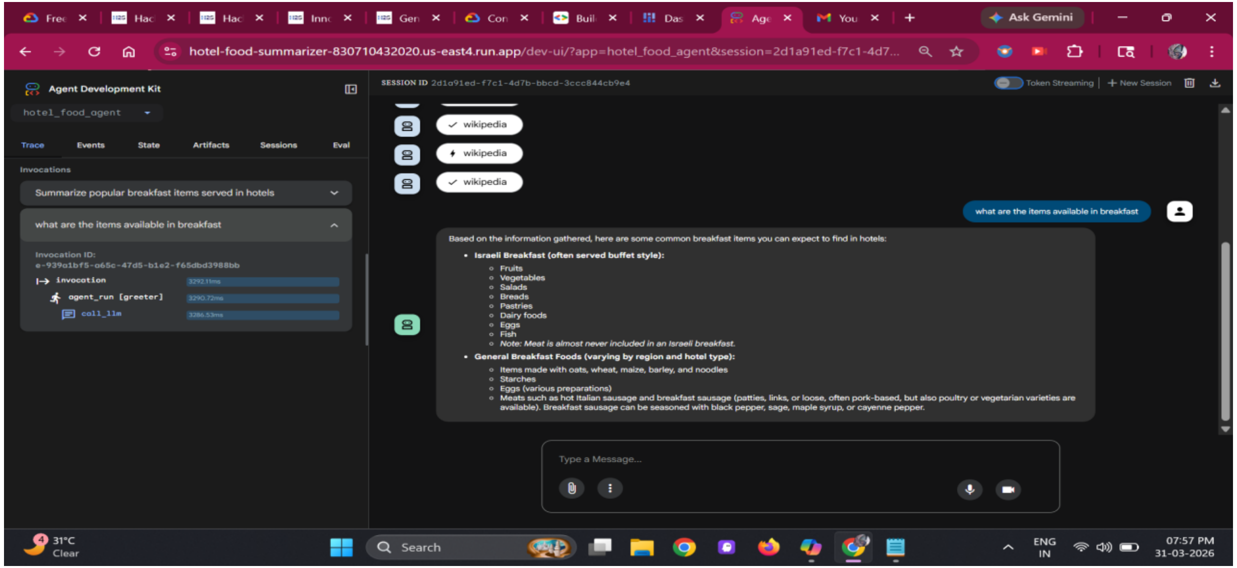

Github url:https://github.com/Yasmeen-Begum/Hack2Skill/tree/main/Gen%20AI%20Academy%20APAC%20Edition/Tracks/Track%201-%20Build%20and%20deploy%20AI%20agents%20using%20Gemini%2C%20ADK%2C%20and%20Cloud%20Run

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Thank you